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kaviya@KV: ~  
GNU nano 8.7.1  
#include <systemc>  
#include <tlm>  
#include <tlm_utils/simple_initiator_socket.h>  
#include <tlm_utils/simple_target_socket.h>  
  
using namespace sc_core;  
using namespace tlm;  
using namespace std;  
  
// Target  
struct Target : sc_module {  
    tlm_utils::simple_target_socket<Target> socket;  
  
    SC_CTOR(Target) : socket("socket") {  
        socket.register_b_transport(this, &Target::b_transport);  
    }  
  
    void b_transport(tlm_generic_payload& trans, sc_time& delay) {  
        int* data = (int*) trans.get_data_ptr();  
        cout << "Target received: " << *data << endl;  
  
        delay += sc_time(10, SC_NS);  
    }  
};  
  
// Initiator  
struct Initiator : sc_module {  
    tlm_utils::simple_initiator_socket<Initiator> socket;  
  
    SC_CTOR(Initiator) : socket("socket") {  
        SC_THREAD(thread_process);  
    }  
  
    void thread_process() {  
        tlm_generic_payload trans;  
        sc_time delay = SC_ZERO_TIME;
```

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// Initiator
struct Initiator : sc_module {
    tlm_utils::simple_initiator_socket<Initiator> socket;

    SC_CTOR(Initiator) : socket("socket") {
        SC_THREAD(thread_process);
    }

    void thread_process() {
        tlm_generic_payload trans;
        sc_time delay = SC_ZERO_TIME;

        int data = 555;

        trans.set_data_ptr(reinterpret_cast<unsigned char*>(&data));
        trans.set_data_length(sizeof(data));

        cout << "Initiator sending: " << data << endl;
        socket->b_transport(trans, delay);

        wait(delay);
    }
};

int sc_main(int argc, char* argv[]) {
    Initiator initiator("initiator");
    Target target("target");

    initiator.socket.bind(target.socket);

    sc_start();

    return 0;
}

kaviya@KV: ~
kaviya@KV:~$ nano exp26.cpp
kaviya@KV:~$ g++ -o exp26 exp26.cpp -lsystemc
kaviya@KV:~$ ./exp26

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Initiator sending: 555
Target received: 555
kaviya@KV:~$ |
```